TextureLab Batch — White Paper

Batch computation of pavement texture descriptors from 3dT scans

Version 3.4.1 · 10 Nov 2026 · Sergio Callai

> The three diagrams of the online version (processing flow, MSD schematic and material ratio curve) cannot be exported to Markdown; they are described in text here. File and field names are in Portuguese, as written by the script.

Summary

`pipeline/texturelab_batch.py` reads 3D pavement scans from the 3dT device (LAZ files) and computes, headless and in batch, the texture descriptors defined in ISO 13473-1, 13473-4, 25178 and 13565-2. Each file produces its own result folder, and the whole batch produces one summary table with one row per measurement point.

- **Profile macrotexture:** MPD and MSD per ISO 13473-1:2019; g-factor per DIN ISO 10844:2024-11 Annex B.
- **Spectrum:** one-third-octave texture levels per ISO 13473-4:2024.
- **Areal parameters:** heights, slopes, material ratio curve and autocorrelation (ISO 25178), in three wavelength bands.
- **Rk family:** core, peaks and valleys of the Abbott curve (ISO 13565-2), for profiles and areas.
- **Hurst exponent and fractal dimension:** descriptive, from the slope of the power spectrum, fitted separately for micro- and macrotexture; from v3.3.0 also in 2D, from the radial PSD of the surface.
- **Height percentiles and 3D g-factor (v3.3.0):** P50/P70/P90/P95 and P90/P70 per areal surface; the Annex B g-factor applied to the surface. Descriptive.
- **Filter verification (v3.4.0):** every filter of the pipeline is checked against the transmission its standard prescribes.

The script was checked against synthetic signals with known answers and against the real file B6 MP1 (SMA 11): MPD = 0.819 mm, in 70 s per file on a single CPU. Microtexture results are flagged as provisional until the optical resolution of the 3dT is known.

1. Scope and objective

The script replaces the calculations of the Streamlit TextureLab app with a single core, checked against the standard texts, that runs headless on a server. Visualisation is done afterwards, from the output files.

Quantity	Standard	Status
MPD, MSD	ISO 13473-1:2019	Conforming (spot measurement)
g-factor and skewness per 100 mm segment	DIN ISO 10844:2024-11, 5.3.2 and Annex B	Conforming
One-third-octave texture level	ISO 13473-4:2024, method 1	Conforming
Areal S parameters (SF and SL5 bands)	ISO 25178-2 and -3, ISO 16610-61 filter	Conforming, with the band choice stated
Areal microtexture parameters	ISO 25178-2 and -3	Provisional
Rk, Rpk, Rvk, Mr1, Mr2 and Sk, Spk, Svk	ISO 13565-2	Conforming algorithm; without the special ISO 13565-1 filter
Rq, Rsk, Rku per segment	—	Descriptive, not ISO 21920
Hurst exponent and fractal dimension (micro and macro)	—	Descriptive, no standard; micro provisional
2D Hurst from the radial PSD of SF (v3.3.0)	—	Descriptive, no standard
Height percentiles P50–P95 and P90/P70 (v3.3.0)	—	Descriptive; read from the ISO 25178-2 material ratio curve
3D g-factor (v3.3.0)	analogue of DIN ISO 10844 Annex B	Descriptive, no standard
Wavelet energies and anisotropy (v3.2.0)	—	Descriptive, no standard

g-factor and skewness (v3.2.0): per DIN ISO 10844:2024-11, clause 5.3.2 and Annex B, per 100 mm segment of the MPD profile (ISO 13473-1 processing without low-pass, zero mean); z_mid = (z_max + z_min)/2 and g = cumulative distribution where z_mid crosses the bearing area curve; mean of the segments per measuring point (`g_factor_media`). The skewness of 5.3.2 is the per-segment Rsk (ISO 13473-2), mean in `perfil_Rsk_media`. Clause 5.3.3 asks for bands from 5 to 100 mm; with ~257 mm scans and l ≥ 12·λ (ISO 13473-4) only bands up to 20 mm are reachable.

ETD: no longer written from v3.2.0 (it is just 1.1 · MPD).

Filter verification (v3.4.0): `tools/verificar_filtros.py` sends sine waves of known wavelength through the same filter functions the pipeline uses and compares the output amplitude with the theoretical transmission. It covers the ISO 16610-61 Gaussian S 0.05 mm, L 0.5 mm and L 5 mm (50 % at λc in both directions), SL5 = SF − G(SF), the chain-A Butterworth low-pass of ISO 13473-1 (2.40 mm, forward and backward) and the third-octave bands of ISO 13473-4 / IEC 61260-1. All pass; report and figure in `docs/verificacao_filtros/`.

ML5 (v3.4.0, removed in v3.4.1): a fourth areal surface (F plane + L 5 mm, no S-filter) was tried and dropped, because it matched SL5 within 0.01 in Ssk and 0.001 mm in Sk. Results from v3.4.0 still carry `ML5_*` columns.

Deliberately out of scope: ENDT (removed from ISO 10844 in the 2021 edition) and the ISO 21920-2 profile parameters.

2. Input data

The input is a folder of 3dT LAZ files, searched recursively. Each file is the complete grid of one measurement point (MP).

Device

Property	Value
Device	3dT-MM (BASt), laser line triangulation, dual 405 nm sensor; originally evaluated with BATex
Measuring field	45 × 300 mm (253 mm after evaluation)
Horizontal resolution used	0.011 mm (adjustable from 0.010 to 1.0 mm)
Vertical resolution	0.001 mm
Triangulation angle	39° (ISO 13473-1 §5.5 recommends ≤ 30°)
Lateral optical resolution	unknown

File format

Item	Value in the files checked (B6 SMA 11 and B248 DSHV)
Format	LAS 1.2, point format 3, compressed (LAZ), about 165–170 MB
Points	89–94 million, complete grid with no gaps
Spacing	exactly 0.011 mm on both axes
Unit	mm (not stated in the LAZ; the original TXT states mm)
Driving direction	file Y axis: always 23,333 points, 256.65 mm
Width	file X axis: 3,826 to 4,032 points, 42.1 to 44.4 mm
Extra fields	intensity, classification, time and RGB are all zero

How the script reads it

1. Reads the points with `laspy` (about 2 to 7 s per file).
2. Checks whether the grid is complete and stored row by row. If so, it builds the matrix by `reshape`, with no interpolation. Otherwise it grids the points by averaging each cell.
3. Assumes mm. If the spacing is below 0.001, it treats it as metres and converts.
4. Takes the driving direction as the **longer axis** of the grid, whatever the axis is called in the file.
5. Parses section, surface type, MP, repeat and date from the file name. For example, `B6xxx_SMA11_MP1_3DT_0-011_NR01_20240807` becomes section B6, surface SMA11,

MP1, NR01, 20240807.

Other formats: the original TXT files (about 2.7 GB each) hold the same content as the LAZ and are not needed. The STL files are **not suitable** for computation: they are simplified meshes with about 2.4 % of the points and lowered peaks.

Drop-outs: the files carry no invalid-reading flags. The script handles missing values if any exist and records the fraction found.

3. Processing

The same grid feeds three independent chains, each with the filters its standard requires. They do not mix: the MPD chain discards microtexture on purpose, and the areal chain does not apply the MPD sharpness normalisation.

Diagram (online version): LAZ file → Reading (mm, road = longer axis) → three parallel chains (Chain A · MPD, ISO 13473-1; Spectrum, ISO 13473-4; Areal, ISO 25178 and 13565-2) → Results (folder per MP, summary CSV).

3.1 Chain A — MPD and MSD (ISO 13473-1:2019)

Measures macrotexture depth. Since the 3dT field is at most 300 mm long, the rules for spot measurements apply.

1. **Strips:** groups 46 neighbouring lines (0.506 mm wide) and averages them into one profile per strip along the driving direction, about 85 profiles per file (Table D.1, note b: 0.5 to 1 mm).
2. **Resampling to 0.5 mm** by averaging the samples in each interval (§7.4). A 256.65 mm profile becomes 513 samples.
3. **Drop-outs:** linear interpolation, extrapolation of up to 5 mm at the ends, and segments with more than 10 % invalid data are discarded (§7.3).
4. **Spikes (Annex E):** flags a sample when $z_i - z_{i-1} \geq 3 \cdot \Delta x$, i.e. a 1.5 mm jump between samples, in both directions. Flagged samples are interpolated. Segments with more than 5 % spikes are discarded.
5. **Low-pass:** 2nd-order Butterworth designed with −3 dB at 2.40 mm, applied forward and backward (effective cut-off 3 mm). The coefficients are identical to Table D.2.
6. **Segments:** two 100 mm segments per profile, centred, with about 28 mm margin at each end to keep away from filter transients.
7. **Slope suppression:** the regression line of each segment is subtracted, the standard's method for spot measurements (§7.6).
8. **MSD:** mean of the peaks of each half of the segment minus the mean level. **MPD:** mean of the valid MSD values, with standard deviation and number of segments (§7.10). ETD (= 1.1 · MPD, the 2019 formula) was written up to v3.1 only.

3.2 Texture spectrum (ISO 13473-4:2024, method 1)

Measures how much amplitude lies in each wavelength range, in one-third octaves.

1. Uses the same 0.5 mm strips, but **without** the MPD filters (ISO 13473-1, §7.1).
2. Applies anti-aliasing (10th-order Butterworth, forward and backward) and reduces the spacing to 0.099 mm.
3. Detects spikes per Annex D, with the same criterion as Annex E.
4. Removes the regression line (Annex F.1) and mirrors the start of the profile to settle the filters (Annex F.2).
5. Filters each one-third-octave band (Butterworth, band edges per IEC 61260-1) and computes the RMS.
6. Converts to a level in dB re 1 µm.

Bands computed: 0.4 mm to 20 mm. The upper limit comes from the rule $l \geq 12 \cdot \lambda max$: with 256.65 mm profiles, the largest valid band is 20 mm.

3.3 Areal parameters (ISO 25178-2 and -3)

Analyses the whole surface with Gaussian filters (ISO 16610-61), in three bands:

Surface	S-filter	F-operation	L-filter	What it represents
SF	0.05 mm	plane	—	All texture above 0.05 mm
SL5	0.05 mm	plane	5 mm	Fine macrotexture and coarse microtexture (0.05 to 5 mm)
MICRO	none (instrument)	plane	0.5 mm	Microtexture (below 0.5 mm), provisional

Why 0.05 mm: under Table 3 of ISO 25178-3 (optical surfaces, 3:1 ratio), a 0.011 mm spacing is only compatible with S-filters from 0.05 mm upwards. The 0.05 / 5 mm pair is a 100:1 combination from Table 1.

Why MICRO is provisional: the L/S ratio is close to 10:1, outside the standard combinations of Table 1, and the optical resolution of the 3dT is unknown.

Edges: the parameters exclude a margin equal to the L-filter on each side (5 mm for SF and SL5; 0.5 mm for MICRO).

For each surface the script computes:

- **Heights:** Sa, Sq, Ssk, Sku, Sp, Sv, Sz.
- **Slope and area:** Sdq (root mean square gradient) and Sdr (developed area increase, in %).
- **Material ratio curve:** Sk, Spk, Svk, Smr1, Smr2, by the ISO 13565-2 method.
- **Volumes:** Vmp, Vmc, Vvc and Vvv, with p = 10 % and q = 80 %.
- **Autocorrelation:** Sal (distance at which the autocorrelation drops to 0.2) and Str (isotropy ratio). For SF and SL5 they are computed on a reduced surface; for

MICRO, on a central 22.5 × 22.5 mm window.
- **Height percentiles (v3.3.0):** P50, P70, P90 and P95 of the heights relative to the mean, read from the same material ratio curve, and the ratio P90/P70 (upper tail of the height distribution).

3.4 Rk family (ISO 13565-2)

The same algorithm serves profiles (Rk) and areas (Sk):

1. Builds the material ratio (Abbott) curve with 10,001 points.
2. Finds the 40 % window with the smallest secant slope.
3. Fits a least-squares line in that window and extends it to 0 % and 100 %, which defines Rk, Mr1 and Mr2.
4. Computes Rpk and Rvk as the heights of triangles with the same area as the peaks and the valleys.

The special ISO 13565-1 filter (valley suppression) is not applied.

3.5 Profile parameters per segment (descriptive)

For each 100 mm segment of chain A the script computes Ra, Rq, Rsk, Rku, Rp, Rv, Rz and the Rk family. It uses the cleaned 0.5 mm profile with the slope suppressed, but **without** the low-pass. These values support comparisons between samples, but do not follow ISO 21920.

3.6 Hurst exponent and fractal dimension (descriptive)

None of the standards used defines these parameters for pavements. The script therefore computes them as descriptive values from the power spectral density (PSD) along the driving direction, and fits the two ranges **separately**: pavement texture is rarely fractal over a wide range.

1. Takes one line every 0.506 mm of width (about 85 lines) at the native 0.011 mm resolution. It does not average lines together, which would attenuate microtexture.
2. Computes each line's PSD by Welch's method: 90 mm Hann window (8,192 samples), 50 % overlap, linear trend removed per segment. Averages the PSD over the lines.
3. In each range, groups the PSD into 20 log-spaced bins and fits a straight line in log-log. The slope gives the exponent β.
4. Converts β to H, and H to the profile and surface fractal dimensions.

Range	Wavelength	Spatial frequency
Micro	0.05 to 0.5 mm	2 to 20 mm^{-1}
Macro	0.5 to 20 mm	0.05 to 2 mm^{-1}

How to read it: H lies between 0 and 1 for a self-affine surface. Values outside that interval mean the PSD does not follow a power law in the range; the script flags them, and the corresponding fractal dimension has no physical meaning. The R^2 of the fit shows how straight the PSD really is in log-log.

Limitations: the surface dimension ($D = 3 - H$) assumes isotropic texture. The micro range is provisional for the same reason as the MICRO surface: it depends on the optical resolution of the 3dT. Sensor noise (9.6 µm standard deviation per the manufacturer) tends to lower H_micro.

3.7 Radial 2D PSD and 2D Hurst (descriptive, v3.3.0)

The 1D Hurst of section 3.6 only looks along the driving direction. The radial PSD uses the whole SF surface:

1. Cuts the surface (without the 5 mm edge margin) into square tiles of 2,048 cells (about 22.5 mm), with 50 % overlap; tiles with missing values are skipped.
2. Removes the mean of each tile, applies a 2D Hann window and computes the 2D power spectrum by FFT. Averages the tiles.
3. Averages the spectrum over rings of equal spatial frequency $q = |(fx, fy)|$, in 60 log-spaced bins, from 1/tile to the Nyquist frequency. The normalisation makes the sum of the PSD times df^2 equal to the height variance (unit mm^4).
4. In the same micro and macro ranges as the 1D Hurst, fits a straight line in log-log; $PSD_2D \propto q^{-(2+2H)}$ gives H, and $D = 3 - H$.

The curve goes to `psd_radial.csv`; the fit to `H2D_micro`, `H2D_macro` and their β, R^2 and D. Same limitations as in 3.6; the 2D fit has the advantage of not depending on the driving direction.

3.8 3D g-factor (descriptive, v3.3.0)

The g-factor of DIN ISO 10844 Annex B is defined on 100 mm profile segments (chain A). The 3D g-factor applies the same rule to the surface, at the same scale, so that it can be compared with the profile value:

1. Averages the SF (and SL5) surface into 0.5 mm × 0.5 mm cells, like the ISO 13473-1 resampling.
2. Cuts it into 100 mm patches along the driving direction, each covering the full width.
3. In each patch, $z_{mid} = (z_{max} + z_{min})/2$ and g = the material ratio at which the bearing area curve crosses z_{mid}, as in Annex B.
4. Reports the mean and standard deviation of the patches (`*_g3D_pct`, `*_g3D_desvio`) and their number.

Each patch has about 85 times more points than one profile segment, so z_{min} is deeper and the 3D g-factor runs above the profile g-factor. On the full surface at native

resolution z_mid would sit on a single extreme point (g ≈ 100 %), which is why the 0.5 mm cells and 100 mm patches are used.

3.9 Wavelets (descriptive, v3.2.0)

Orthonormal discrete wavelet transform (Daubechies db4, periodisation); the energy of the coefficients equals the variance.

- **Profile:** native-resolution lines (one per 0.5 mm strip); energy per octave $\lambda \in [2^j \cdot dx, 2^{j+1} \cdot dx]$, in µm rms and dB re 1 µm, and the micro (< 0.5 mm) and macro shares (`ondaletas_perfil.csv`, `W_rms_micro_um`, `W_rms_macro_um`, `W_fracao_micro`).
- **2D:** on the SL5 surface, energy per octave along the road, across it and diagonally; anisotropy = (E_road − E_across)/(E_road + E_across) (`ondaletas_2d.csv`, `W2_anisotropia_micro`, `W2_anisotropia_macro`).

Per octave, so not comparable 1:1 with the third-octave spectrum (one octave holds three bands, ≈ +4.8 dB).

4. Outputs

Each LAZ file gets its own folder. At the end of the batch, the script merges all summaries into one table with one row per MP. Raw data are never modified.

```
results/
  execucao.json                configuration, versions, date, duration
  resumo_geral.csv             one row per file, all parameters
  B6/
    B6xxx_SMA11_MP1_3DT_0-011_NR01_20240807/
      resumo.json
      cadeiaA_segmentos.csv
      espectro_terco_oitava.csv
      espectro_por_perfil.csv
      psd_media.csv
      psd_radial.csv           from v3.3.0
      ondaletas_perfil.csv     from v3.2.0
      ondaletas_2d.csv         from v3.2.0
      previa.npz
      visualizacao.npz         curves for the viewer
      erro.txt                 only if the file fails
  B248/
    ...
```

4.1 Files

File	Content	Typical use
`resumo_geral.csv`	One row per MP, with every field of `resumo.json`	Compare sections and surfaces; basis for charts
`resumo.json`	All scalar results of one MP, metadata and timings	Inspect one MP; audit
`cadeiaA_segmentos.csv`	One row per 100 mm segment: strip, position, MSD, validity, spike and drop-out fractions, profile parameters	MSD variability; MSD across the width
`espectro_terco_oitava.csv`	One row per band: centre wavelength, mean level in dB, standard deviation, number of profiles	Spectrum chart of the MP
`espectro_por_perfil.csv`	Level of each band (column) for each profile (row)	Spectrum scatter; uncertainty
`psd_media.csv`	Mean PSD along the driving direction: frequency (mm^{-1}), wavelength (mm) and PSD (mm^3)	Visual check of the Hurst fit; log-log chart
`psd_radial.csv`	Radial 2D PSD of SF: spatial frequency q (mm^{-1}), wavelength (mm), PSD_2D (mm^4) and number of modes per bin (v3.3.0)	2D Hurst check; log-log chart
`ondaletas_perfil.csv`, `ondaletas_2d.csv`	Wavelet energy per octave: profile (µm rms, dB) and 2D directional energies on SL5 (v3.2.0)	Micro/macro shares; anisotropy
`previa.npz`	Height map reduced 8× on each axis (about 490 × 2,900 cells, float16) and its spacing in mm	Light 2D/3D visualisation
`visualizacao.npz`	Abbott curves, height histograms, example profiles and segments for the viewer	Viewer only; does not change the parameters
`execucao.json`	Full configuration, script version, Python, numpy, platform, date, duration and number of processes	Reproducibility

4.2 Summary fields

Heights and volumes are in mm (volumes in mm^3/mm^2, which equals mm; multiply by 1,000 for ml/m^2). Lengths are in mm and material ratios in %.

Group	Fields	Meaning
Identification	`arquivo`, `trecho`, `revestimento`, `mp`, `nr`, `data`, `caminho`	File name, section, surface, MP, repeat, date and path
Reading	`n_pontos`, `n_largura`, `n_via`, `dx_mm`, `largura_mm`, `comprimento_mm`, `leitura`, `frac_invalidos`, warnings	Geometry read; `leitura` = `reshape` (fast) or `generica`
Chain A	`MPD`, `MPD_desvio`	Mean profile depth and its standard deviation across segments (mm); ETD = 1.1·MPD up to v3.1
ISO 10844	`g_factor_media`, `g_factor_desvio`, `g_factor_n_segmentos`; `g_pct` per segment	Shape factor (%) per 100 mm segment, mean and std per measuring point (v3.2.0)
Chain A, control	`A_n_faixas`, `A_n_segmentos_total`, `A_n_segmentos_validos`, `A_largura_faixa_mm`, `A_inicio_primeiro_segmento_mm`, `A_spike_frac_media`, `A_dropout_frac_media`	How many profiles and segments entered the MPD and how much was corrected
Profile (descriptive)	`perfil_Rq_media`, `perfil_Rsk_media`, `perfil_Rku_media`, `perfil_Rk_media`, `perfil_Rpk_media`, `perfil_Rvk_media`, `perfil_Rmr1_media`, `perfil_Rmr2_media`	Mean over valid segments
Spectrum	`E_passo_mm`, `E_comprimento_avaliacao_mm`, `E_n_perfis`, `E_lambda_min_mm`, `E_lambda_max_mm`, `E_spike_frac_media`	Analysis conditions; the levels are in the spectrum CSV files
Areal, per surface (`SF_`, `SL5_`, `MICRO_`)	`Sa`, `Sq`, `Ssk`, `Sku`, `Sp`, `Sv`, `Sz`	Heights: mean absolute, RMS, skewness, kurtosis, highest peak, deepest valley and total range
	`Sdq`, `Sdr_pct`	RMS slope (dimensionless) and real area increase over the projected area (%)
	`Sk`, `Spk`, `Svk`, `Smr1`, `Smr2`	Core, reduced peaks and reduced valleys of the material ratio curve
	`Vmp`, `Vmc`, `Vvc`, `Vvv`	Material volume of peaks and core; void volume of core and valleys
	`Sal`, `Str`, `acf_aviso`	Autocorrelation length, isotropy ratio (0 = directional, 1 = isotropic) and a warning if the window limits the computation
	`P50`, `P70`, `P90`, `P95`, `P90_P70`	Height percentiles relative to the mean (mm) and the ratio P90/P70 (v3.3.0)
	`g3D_pct`, `g3D_desvio`, `g3D_n_trechos` (SF and SL5 only)	3D g-factor (%): mean and std over the 100 mm patches, and their number (v3.3.0)

Group	Fields	Meaning
	`MICRO_status`	Text marking the microtexture as provisional
Hurst and fractal (suffix `_micro` or `_macro`)	`H`, `D_perfil`, `D_superficie`, `H_beta`, `H_R2`, `H_faixa_mm`, `H_aviso`, `H_n_linhas`, `H_segmento_mm`, `H_micro_status`	Hurst exponent, profile and surface fractal dimension, PSD exponent, fit quality, range used and a warning if H falls outside (0, 1)
2D Hurst (suffix `_micro` or `_macro`)	`H2D`, `D2D_superficie`, `H2D_beta`, `H2D_R2`, `H2D_aviso`; `R_n_blocos`, `R_bloco_mm`	2D Hurst exponent, surface fractal dimension, PSD₂D exponent, fit quality; number and size of the tiles (v3.3.0)
Wavelets	`W_rms_micro_um`, `W_rms_macro_um`, `W_fracao_micro`, `W2_anisotropia_micro`, `W2_anisotropia_macro`	Micro and macro wavelet energy (µm rms), micro share, 2D anisotropy (v3.2.0)
Run	`status`, `erro`, `versao_script`, `leitura_s`, `cadeia_A_s`, `espectro_s`, `hurst_s`, `areal_s`, `total_s`	Outcome and time of each step (s)

4.3 Reading the results

- **MPD** is the number to compare with other devices and with the sand patch (ETD = 1.1 · MPD if needed). The MPD of a site is the mean of its MPs, reported with the standard deviation and the number of segments.
- **The spectrum** shows at which wavelengths the texture energy lies. It is the basis for tyre/road noise studies.
- **Negative Ssk** means a surface dominated by voids (negative texture, typical of SMA). Positive Ssk means a surface dominated by peaks.
- **SF** describes the whole texture, **SL5** isolates the scale of fine aggregate and **MICRO** describes aggregate harshness, still as a provisional value.

4.4 Parameter equations

These are the formulas the script computes, in the order of the groups in table 4.2. For continuous formulas the script uses the discrete form: integrals become means over the grid samples.

Resampling (chain A and spectrum). Each new sample is the mean of the original samples within the interval Δ (0.5 mm in chain A):

$$\bar{z}_k = \frac{1}{N_k} \sum_{i \in I_k} z_i, \qquad I_k = \{\, i : k\Delta \le x_i < (k+1)\Delta \,\}$$

Spikes (ISO 13473-1 Annex E; ISO 13473-4 Annex D). A sample is flagged as invalid when it rises more than α · Δx above its neighbour, in either direction, and is then interpolated:

$$z_i - z_{i-1} \geq \alpha\,\Delta x \quad \text{or} \quad z_i - z_{i+1} \geq \alpha\,\Delta x, \qquad \alpha = 3$$

Chain A low-pass. 2nd-order Butterworth applied forward and backward, with a 2.40 mm design λc. The recursion is Formula D.1 of the standard; the resulting magnitude is the square of a single pass:

$$y_i = \frac{x_i + 2x_{i-1} + x_{i-2}}{A_0} + A_1\,y_{i-2} + A_2\,y_{i-1}$$
$$|H(\lambda)| = \frac{1}{1 + (\lambda_c/\lambda)^4}, \qquad \lambda_c = 2.40\ \text{mm}$$

Slope suppression, MSD and MPD (ISO 13473-1). In each 100 mm segment the least-squares line is subtracted; MSD compares the peaks of both halves with the mean level:

$$z'(x) = z(x) - (a + b\,x)$$
$$MSD = \frac{\max_{\text{1st half}} z' + \max_{\text{2nd half}} z'}{2} - \overline{z'}$$
$$MPD = \frac{1}{N}\sum_{j=1}^{N} MSD_j$$

MSD schematic (online version): the highest peak is searched separately in each half of the segment, after slope suppression; MSD is the vertical distance between the mean of those two peaks and the mean level of the segment.

Spectrum (ISO 13473-4). Each one-third-octave band has centre $f_m = 10\text{^}(n/10)$ m^{-1} and edges at ±1/6 octave (base 10). The profile is extended at the start by mirroring (Annex F), band-filtered and converted to a level:

$$f_1 = f_m \cdot 10^{-1/20}, \qquad f_2 = f_m \cdot 10^{1/20}$$
$$z_{-k} = 2z_0 - z_k$$
$$a_\lambda = \sqrt{\frac{1}{L}\int_0^L y_\lambda^2(x)\,dx}, \qquad L_{tx,\lambda} = 20\log_{10}\frac{a_\lambda}{10^{-6}\ \text{m}}$$

Areal Gaussian filter (ISO 16610-61). Weighting function and transmission, which is 50 % at λ = λc; the S-filter smooths, the L-filter removes the smoothed part:

$$s(x,y) = \frac{1}{\alpha^2\lambda_c^2}\exp\!\left(-\pi\frac{x^2+y^2}{\alpha^2\lambda_c^2}\right), \qquad \alpha = \sqrt{\ln 2/\pi} \approx 0.4697$$
$$H(\lambda) = \exp\left[-\pi\left(\frac{\alpha\lambda_c}{\lambda}\right)^2\right]$$
$$Z_{SF} = (s_{N_{is}} * Z) - \text{plane}, \qquad Z_{SL} = Z_{SF} - s_L * Z_{SF}$$

Heights (ISO 25178-2). On the filtered surface, with zero mean and evaluation area A:

$$S_a = \frac{1}{A}\iint_A |z|\,dA, \qquad S_q = \sqrt{\frac{1}{A}\iint_A z^2\,dA}$$
$$S_{sk} = \frac{1}{S_q^3}\,\frac{1}{A}\iint_A z^3\,dA, \qquad S_{ku} = \frac{1}{S_q^4}\,\frac{1}{A}\iint_A z^4\,dA$$
$$S_p = \max z, \qquad S_v = |\min z|, \qquad S_z = S_p + S_v$$

Slope and developed area. Derivatives are central differences on the grid:

$$S_{dq} = \sqrt{\frac{1}{A}\iint_A \left[\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2\right] dA}$$
$$S_{dr} = \frac{100\,\%}{A}\iint_A \left(\sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2} - 1\right) dA$$

Material ratio curve and Rk / Sk family (ISO 13565-2). c(Mr) is the height above which a fraction Mr of the surface lies. In the 40 % window with the smallest slope, the line $\ell(\mathrm{Mr}) = \mathrm{k}\cdot\mathrm{Mr} + \mathrm{q}$ is fitted:

$$R_k = \ell(0) - \ell(100), \qquad c(Mr_1) = \ell(0), \qquad c(Mr_2) = \ell(100)$$
$$A_1 = \int_0^{Mr_1} [c(Mr) - \ell(0)]\,dMr, \qquad R_{pk} = \frac{2A_1}{Mr_1}$$
$$A_2 = \int_{Mr_2}^{100} [\ell(100) - c(Mr)]\,dMr, \qquad R_{vk} = \frac{2A_2}{100 - Mr_2}$$

Material ratio curve schematic (online version): the line fitted in the 40 % window with the smallest slope, extended to 0 % and 100 %, bounds the Rk core; the triangles have the same area as the peaks above and the valleys below the core, and their heights are Rpk and Rvk.

Volumes (ISO 25178-2), with p = 10 % and q = 80 %:

$$V_m(p) = \frac{1}{100}\int_0^p [c(Mr) - c(p)]\,dMr, \qquad V_v(p) = \frac{1}{100}\int_p^{100} [c(p) - c(Mr)]\,dMr$$
$$V_{mp} = V_m(10), \quad V_{mc} = V_m(80) - V_m(10), \quad V_{vc} = V_v(10) - V_v(80), \quad V_{vv} = V_v(80)$$

Autocorrelation, Sal and Str (ISO 25178-2). The autocorrelation is computed by FFT with zero-padding. Sal is the shortest shift at which it drops to 0.2; Str divides that by the longest:

$$f_{ACF}(\tau_x,\tau_y)=\frac{\iint z(x,y)\,z(x+\tau_x,y+\tau_y)\,dx\,dy}{\iint z^2(x,y)\,dx\,dy}$$

$$S_{al}=\min_{f_{ACF}(\tau)\le 0.2}\|\tau\|,\qquad S_{tr}=\frac{S_{al}}{\max_\theta\ \tau_{0.2}(\theta)}$$

Hurst exponent and fractal dimension (descriptive). In each range, the mean Welch PSD is fitted by a power law in log-log:

$$PSD(f)\propto f^{-\beta},\qquad \beta=1+2H\quad\Rightarrow\quad H=\frac{\beta-1}{2}$$

$$D_{\text{profile}}=2-H,\qquad D_{\text{surface}}=3-H$$

Radial 2D PSD and 2D Hurst (descriptive, v3.3.0). For each tile of side t (t × t cells, spacing Δx) with the 2D Hann window w, and the radial mean over the ring of frequency q:

$$C(f_x,f_y)=\frac{\Delta x^2}{\sum w^2}\left\langle\left|\mathcal{F}\{w\cdot(z-\bar z)\}\right|^2\right\rangle_{\text{tiles}},\qquad \sum C\,\Delta f^2=\sigma_z^2$$

$$C_{2D}(q)=\langle C(f_x,f_y)\rangle_{|(f_x,f_y)|\approx q}\propto q^{-(2+2H)}\quad\Rightarrow\quad H=\frac{\beta-2}{2},\qquad D=3-H$$

Height percentiles (v3.3.0). The p-th percentile is the material ratio curve at Mr = 100 − p, relative to the mean height:

$$P_p=c(100-p)-\bar z,\qquad p\in\{50,70,90,95\},\qquad \text{ratio } P_{90}/P_{70}$$

g-factor (DIN ISO 10844 Annex B) and 3D g-factor. In each 100 mm segment (profile) or 100 mm patch of 0.5 mm cells (surface), with Mr in % counted from the highest point:

$$z_{\text{mid}}=\frac{z_{\max}+z_{\min}}{2},\qquad g=Mr\ \text{ such that }\ c(Mr)=z_{\text{mid}},\qquad g_{3D}=\frac{1}{N}\sum_{j=1}^{N}g_j$$

5. How to run

The script is a single Python file. It runs headless and without a GPU.

Requirements: Python 3.11 or later and about 4 GB of RAM per file processed in parallel. On the shared server use `pipeline/servidor.sh` (own venv, CPU/RAM/disk limits, pause and resume); see `pipeline/README.md`.

```
pip install -r pipeline/requirements.txt
```

Test with one file (recommended before the batch):

```
python pipeline/texturelab_batch.py --input /path/to/laz --output /path/to/results --
pattern "*B6*MP1*.laz"
```

Full batch, inside a `tmux` session so it keeps running if the SSH connection drops:

```
tmux new -s texture
python pipeline/texturelab_batch.py --input /path/to/laz --output /path/to/results
```

Option	Effect
`--input`	Folder with the LAZ files, searched recursively (required)
`--output`	Results folder, created if missing (required)
`--pattern`	Name filter, e.g. `"*B248*.laz"` (default: `*.laz`)
`--workers`	Number of files in parallel (default: automatic, from cores and RAM)
`--skip-done`	Skips files that already have a result with status `ok`, to resume an interrupted batch

While running, the script prints one line per finished file, with its MPD and time. A failing file does not stop the batch: the error goes to `erro.txt` in the file's folder and to the `erro` field of the summary.

The normative settings (strip width, filters, bands) live in the `CFG` dictionary at the top of the script and are saved to `execucao.json` on every run.

6. Normative references

Standard	Short title	Used for
ISO 13473-1:2019 (DIN EN ISO 13473-1:2021-11)	Mean profile depth	Chain A: MPD, MSD; Annexes D and E
ISO 13473-2:2002 (DIN ISO 13473-2:2004-07)	Terminology and requirements for profile analysis	One-third-octave bands; micro and macro ranges
ISO 13473-4:2024	Spectral analysis of surface profiles	Spectrum: method 1, Annexes D and F
ISO 13473-5 (DIN EN ISO 13473-5:2024-05)	Megatexture	Definition of microtexture (< 0.5 mm)
ISO 25178-2	Areal parameters	Definition of the S and V parameters
ISO 25178-3:2012 (DIN EN ISO 25178-3:2012-11)	Specification operators	S-F-L order; Tables 1 and 3 (choice of S and L)
ISO 16610-61	Areal Gaussian filter	S and L filters
ISO 13565-2:1996 (DIN EN ISO 13565-2:1998-04)	Material ratio curve: Rk	Rk and Sk family
IEC 61260-1	Octave-band filters	One-third-octave band edges
DIN ISO 10844:2024-11	Test tracks for tyre/road noise	g-factor and skewness (5.3.2, Annex B); 3D g-factor by analogy
ISO 21920-2, ISO 13565-1	—	Not applied